Environmental Research 261 (2024) 119706



Contents lists available at ScienceDirect

Environmental Research

journal homepage: www.elsevier.com/locate/envres





Association between prenatal air pollutant exposure and autism spectrum disorders in young children: A matched case-control study in Canada

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ARTICLE INFO

Keywords:
Air pollution
Autism spectrum disorder
Matched case-control
Odds ratio
Prenatal exposure
Time-varying effects

ABSTRACT

The direction and magnitude of association between maternal exposure to ambient air pollutants across gestational windows and offspring risk of autism spectrum disorders (ASD) remains unclear. We sought to evaluate the time-varying effects of prenatal air pollutant exposure on ASD. We conducted a matched case-control study of singleton term children born in Ontario, Canada from 1-Apr-2012 to 31-Dec-2016. Provincial birth registry data were linked with applied behavioural analysis services and ambient air pollutant datasets to ascertain prenatal exposure to nitrogen dioxide (NO_2), ground-level ozone (O_3), fine particulate matter ($PM_{2.5}$), and ASD diagnoses. Covariate balance between cases and controls was established using coarsened exact matching. Conditional logistic regression was used to assess the association between prenatal air pollutant exposure and ASD. Distributed lag non-linear models (DLNM) were used to examine the effects of single-pollutant exposure by prenatal week. Sensitivity analyses were conducted to assess the impact of exposure period on the observed findings. The final sample included 1589 ASD cases and 7563 controls. Compared to controls, cases were more likely to be born to mothers living in urban areas, delivered by Caesarean section, and assigned male sex at birth. NO_2 was a consistent and significant contributor to ASD risk after accounting for co-exposure to O_3, $PM_{2.5}$ and covariates. The odds ratio per interquartile range increase was 2.1 (95%CI 1.8–2.3) pre-conception, 2.2 (2.0–2.5) for the 1st trimester, 2.2 (1.9–2.5) for the 2nd trimester, and 2.1 (1.9–2.4) for the 3rd trimester. In contrast, findings for O_3 and $PM_{2.5}$ with ASD were inconsistent. Findings from DLNM and sensitivity analyses were similar. Exposure to NO_2 before and during pregnancy was significantly associated with ASD in offspring. The relationship between prenatal O_3 and $PM_{2.5}$ exposure and ASD remains unclear. Further investigation into the combined effects of multi-pollutant exposure on child neurodevelopment is warranted.

1. Introduction

Autism spectrum disorders (ASD) is a neurological and developmental disorder that can have profound impacts on affected persons, their families and communities. ASD is characterized by symptoms such as presence of repetitive behaviours or restricted interests, and impairments in communication and social interaction (Brian et al., n.d.; Hyman et al., 2020). ASD is described as a "developmental disorder", as the symptoms are generally observed from age 18–24 months, but it can be diagnosed at any age (National Institute of Mental Health, 2024). Presentation can range from mild without intellectual disability to severe with significant impairments and comorbid conditions (Hyman

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https://doi.org/10.1016/j.envres.2024.119706
Received 4 June 2024; Received in revised form 22 July 2024; Accepted 28 July 2024
Available online 30 July 2024
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et al., 2020). Reported prevalence estimates generally range from 1 to 2% across jurisdictions but are influenced greatly by the data source (e.g., health administrative data, patient registries, parent reported surveys), physician coding practices and health system structure (e.g., publicly funded, privately funded), diagnostic criteria and year of study (Chiarotti and Venerosi, 2020; Lyall et al., 2017). The United States' Centers for Disease Control estimates that 18.5 per 1000 children (1 in 54) age 8 years were affected by ASD in 2016 (Maenner et al., 2021). In Canada, the prevalence of ASD among children 5–17 years of age was 1.52% for the 2014–2015 fiscal year (Public Health Agency of Canada, 2018). Increases in prevalence of ASD over the last decade are noted, and may be attributable to increases in diagnoses of milder cases of ASD, changes in diagnostic criteria and community awareness (Chiarotti and Venerosi, 2020).

ASD etiology remains poorly understood but is likely multifactorial, with both genetic and environmental contributions. A recent Canadian study reported both null and protective effects of exposures to various greenspace metrics such as active living environments and proximity to parks on ASD (Lavigne et al., 2024). In contrast, prenatal air pollutant exposure has emerged as a risk factor of particular interest (Lam et al., 2016; Flores-Pajot et al., 2016; Chun et al., 2020). Alterations in placental development, increased oxidative stress and epigenetic changes are demonstrated among mothers exposed to high air pollutant concentrations (Saenen et al., 2019a; van den Hooven et al., 2012a) and accumulating evidence suggests that exposure may be associated with a range of adverse health outcomes including preterm birth, low birth weight, infant mortality and respiratory conditions in childhood (Hehua et al., 2017; Johnson et al., 2022; Jung et al., 2020; Guan et al., 2019; Stieb et al., 2012). Air pollution is a mixture of various components known to have harmful effects on human health and the environment, and for which governments have set air quality standards (Environment Canada, 2017; United States Environmental Protection Agency, 2021). Major sources include fuel combustion and power generation, which emit nitrogen oxides (NO_x), components of particular matter (PM_x), and organic and inorganic compounds that react in the atmosphere to produce additional pollutant including ground-level ozone (O_3) and secondary fine particulate matter ($PM_{2.5}$). While some pollutants such as sulfur dioxide (SO_2) and carbon monoxide (CO) are ubiquitous, data suggest that nitrogen dioxide (NO_2), O_3 and $PM_{2.5}$ in particular may have a substantial impact on mortality and morbidity for non-contagious (or chronic) diseases such as cancer, cardiovascular, respiratory disease (Health Canada, 2021).

Findings from systematic reviews and meta-analyses suggest weak to moderate evidence for associations between ASD and prenatal exposure to NO_x, O_3, and PM_x (Lam et al., 2016; Flores-Pajot et al., 2016; Chun et al., 2020). Differences in study design, case ascertainment, exposure windows, and contaminants of interest hamper robust comparison across studies, however. Furthermore, published data are frequently limited by a lack of data on maternal sociodemographic information, pre-existing medical history and obstetrical characteristics that may be relevant to both the exposure and outcome. To study environmental effect on ASD in Canada, we built a retrospective cohort of young children with or without ASD, for which we conducted two separated investigations on greenspace (Lavigne et al., 2024) and ambient air pollution (this study). In this study, we investigated adverse health effects of maternal air pollution exposure on ASD in offspring by exposure time windows: before and during pregnancy by gestational age in week and trimester while adjusting for a number of relevant confounders. We sought to address whether prenatal exposure to ambient air pollution was associated with ASD in young children (<6 years old), and whether there were critical exposure time windows during which the fetus may be particularly vulnerable to environmental toxicants.

2. Materials and methods

2.1. Study design

The target population of this study was young children at early childhood years (ages 2–6 years) with or without ASD, and the sample population was those in Ontario, covering about 40% of Canada children. We conducted a matched case-control study of full-term singleton children (≥37 weeks gestation) born in Ontario between 1-April-2012 and 31-December-2016. We applied a Coarsened Exact Matching (CEM) method to generate a cohort of matched case-control children with matching ratios ranged from 1:1 to 1:5 case to controls. Matching variables consisted of mainly maternal characteristics and one infant characteristic: maternal age at delivery, race/ethnicity, parity prenatal care provider, substance use, pre-existing health conditions, and mental health at personal level, neighbourhood education and income, and infant birth year, (Table 1). The study period was determined by data availability from the data sources described below. The follow-up period for each child included pre-pregnancy (13 weeks prior to conception) and pregnancy (conception to birth) period.

2.2. Data sources and linkage

Data were derived from three databases of children with ASD, mothers, and air pollution concentrations: (1) the Ontario Ministry of Children and Youth Services' Child and Adolescent Needs and Strengths – Autism Spectrum Profile (CANS-ASP), (2) the Better Outcomes Registry & Network (BORN) Ontario, and (3) the Canadian Urban Environmental Health Research Consortium (CANUE). The CANS-ASP is designed for assessment of autistic individuals based on over 80 criteria to provide the Applied Behaviour Analysis (ABA) services on autistic children (John Praed Foundation, 2021). BORN Ontario is a prescribed birth registry in Ontario, which routinely captures maternal data such as demographics, reproductive history, pregnancy, labour and delivery, newborn screening and newborn care (Murphy et al., 2021). CANUE is a research platform for environmental measures by postal code, including metrics of air quality, green and blue spaces, weather and climate indicators, and neighbourhood-level socioeconomic conditions across Canada ("CANUE: The Canadian Urban Environmental Health Research Consortium," n.d.). CANUE data were derived via ground and satellite monitoring, gridded surface datasets, chemical transport models, and land use regression models. For data linkages of the three databases, we first linked CANS-ASP and BORN Ontario were first linked based on deterministic and probabilistic linkage using several key factors (infant record IDs, sex and date of birth, and maternal name and residence postal at delivery. No missing data were imputed, which resulted in considerable rate of exclusion. We then linked the CANUE datasets using maternal postal code (Fig. 1).

2.3. Health outcome (cases and controls)

The main health outcome of this study is ASD in children diagnosed after 2 years of age (Canadian Paediatric Society, n.d.). Specifically, cases were the autistic children with a record in the CANS-ASP, diagnosed by health professionals (e.g., psychiatrist or psychologist), whereas controls were the children without ASD. Birth records for both cases and controls were available from BORN Ontario, integrating the Children's Hospital of Eastern Ontario (CHEO) Autism Program and CANS-ASP data from 13 agencies (ABA services and support programs) in Ontario ("History - BORN Ontario," 2022; "Ontario Autism Program Clinical Framework | Ontario.ca," n.d.).

2.4. Air pollutants and exposure period

The air pollutants of interest were ambient concentrations of gaseous pollutants (NO_2("CanMap Postal Code Suite," 2015; Hystad et al., 2011;

Table 1
Maternal and infant characteristics at delivery.

	Cases (ASD) N = 1589	Controls N = 7563	Standardized mean difference
Maternal characteristics at delivery			
Maternal age, years			
Mean ± SD	31.2 ± 5.4	31.2 ± 5.3	0.02
Median (Q1-Q3)	31 (27–35)	31 (28–35)	0.02
Maternal age, n(%)			
<20 years	21 (1)	93 (1)	0.01
20–24 years	160 (10)	750 (10)	0.01
25–29 years	401 (25)	1914 (25)	0
30–34 years	564 (36)	2716 (36)	0.01
35–39 years	336 (21)	1697 (22)	0.03
40+ years	107 (7)	393 (5)	0.06
Neighbourhood education quartile, n(%)			
1 (lowest)	300 (19)	1411 (19)	0.01
2	408 (26)	1957 (26)	0
3	551 (35)	2626 (35)	0
4 (highest)	323 (20)	1544 (20)	0
Missing	7 (<1)	25 (<1)	0.02
Neighbourhood household income quartile, n(%)			
1 (lowest)	776 (49)	3685 (49)	0
2	369 (23)	1761 (23)	0
3	257 (16)	1244 (16)	0.01
4 (highest)	176 (11)	841 (11)	0
Missing	11 (<1)	32 (<1)	0.04
Type of Residential Community, n(%)			
Urban	1554 (98)	7089 (94)	0.2
Rural	35 (2)	474 (6)	0.2
Maternal race/ethnicity, n(%)			
White	429 (27)	2044 (27)	0.01
Asian	537 (34)	2589 (34)	0.02
Black	158 (10)	726 (10)	0.01
Other/unknown	093 (6)	388 (5)	0.03
Missing	372 (23)	1816 (24)	0.01
Parity, n(%)			
0	811 (51)	3910 (52)	0.01
1	479 (30)	2296 (30)	0
2	176 (11)	813 (11)	0.01
≥3	92 (6)	422 (6)	0.01
Missing	31 (2)	122 (2)	0.03
Substance use[a], n(%)			
Yes	115 (7)	485 (6)	0.04
No	1345 (85)	6542 (87)	0.04
Missing	129 (8)	536 (7)	0.04
Maternal pre-existing health conditions[b], n(%)			
Yes	73 (5)	252 (3)	0.06
No	1516 (95)	7311 (97)	0.06
Maternal mental health conditions[c], n(%)			
Yes	166 (10)	711 (9)	0.03
No	1423 (90)	6852 (91)	0.03
Antenatal health care provider, n(%)			
Yes	1528 (96)	7329 (97)	0.04
No/unknown	<6 (S)	6 (<1)	0.04
Missing	57 (3.6)	228 (3)	0.03
Infant characteristics at birth			
Birth Year, n(%)			
2012	410 (26)	1950 (26)	0
2013	502 (32)	2371 (31)	0.01
2014	359 (23)	1705 (23)	0
2015	233 (15)	1128 (15)	0.01
2016	85 (5)	409 (5)	0
Birth Season, n(%)			
Spring (Mar–May)	373 (24)	1707 (23)	0.02
Summer (June–Aug)	432 (27)	2228 (30)	0.05
Fall (Sept–Nov)	462 (29)	2146 (28)	0.02
Winter (Dec–Feb)	322 (20)	1482 (20)	0.02
Mode of delivery, n(%)			
Vaginal	1068 (67)	5416 (72)	0.1
Caesarean Section	521 (33)	2147 (28)	0.1
Sex, n(%)			
Male	1247 (79)	3898 (52)	0.59
Female	342 (21)	3662 (48)	0.59
Missing	0 (0.0)	<6 (S)	0.03
Birthweight, grams			
Mean ± SD	3412.7 ± 489.8	3384.4 ± 466.6	0.06
Median (Q1-Q3)	3394 (3096–3710)	3365 (3080–3680)	0.06
Gestational age at birth, weeks			
Mean ± SD	39.1 ± 1.2	39.2 ± 1.2	0.06
Median (Q1-Q3)	39 (38–40)	39 (38–40)	0.06

S, small cell size <6 suppressed; SD, standard deviation; Q1, 25th percentile or lower quartile; Q3, 75th percentile or upper quartile.

[a] a composite of maternal self-reported cocaine, gas/glue, hallucinogens, methadone, narcotics, opioids, cannabis, and other drug use, alcohol use and smoking in pregnancy, alcohol use and smoking in pregnancy.

[b] a composite of maternal pre-existing hypertension, diabetes, asthma and heart disease.

[c] a composite of maternal self-reported anxiety and depression for the index pregnancy.

Weichenthal et al., 2017), O_3(Air Quality Research Division, 2017; Robichaud et al., 2016; Robichaud and Menard, 2014),) and fine particulate matter ($PM_{2.5}$) ("CanMap Postal Code Suite," 2015; Hammer et al., 2020). Weekly exposures were estimated at the postal-code level, the highest temporal-spatial resolution available from the CANUE. To identify critical exposure timing, six time windows were considered: (Johnson et al., 2022) the pre-conception period (from 13 weeks pre-conception to delivery, T0), each trimester of pregnancy (1st trimester, <13 weeks gestation, T1; 2nd trimester, 13 to <27 weeks gestation, T2; 3rd trimester, ≥27 weeks gestation, T3), the entire gestational period from conception to delivery (T4), and the entire study period from 13 weeks pre-conception to delivery (T5). Weekly average exposures were aggregated over each time window. Finally, continuous weekly exposures (T6) over the entire prenatal period were also examined.

2.4.1. Maternal and infant covariates

Potential covariates were identified through a combination of literature reviews, review of a directed acyclic graph (DAG), as well as data availability. Baseline characteristics and covariates were obtained from BORN Ontario. Maternal covariates included age group, neighbourhood education-level quartile, neighbourhood household income quartile, parity, self-reported health information during pregnancy: (1) substance use, a composite of cocaine, gas/glue, hallucinogens, methadone, narcotics, opioids, cannabis, alcohol, smoking, and other drug use; (2) pre-existing health conditions, a composite of circulatory (e.g., hypertension, heart disease), respiratory (e.g., asthma), and other chronic illness (e.g., diabetes); mental health conditions, a composite of maternal anxiety and depression; and type of residential community at delivery (e.g., urban, rural). Infant covariates were information at birth such as gestational age, mode of delivery, sex, weight, year, and season.

2.5. Statistical analysis

Summary statistics and standardized mean differences were used to describe the study cohort and compare maternal and infant baseline characteristics between ASD cases and controls. Basic statistics were used to describe the data: means with standard deviations (SD) or medians, and interquartile ranges (IQR) for continuous variables; and frequencies (n) and percentages (%) for categorical variables. To check covariate balance between cases and controls, a standardized mean difference was considered, indicating covariate imbalance if the

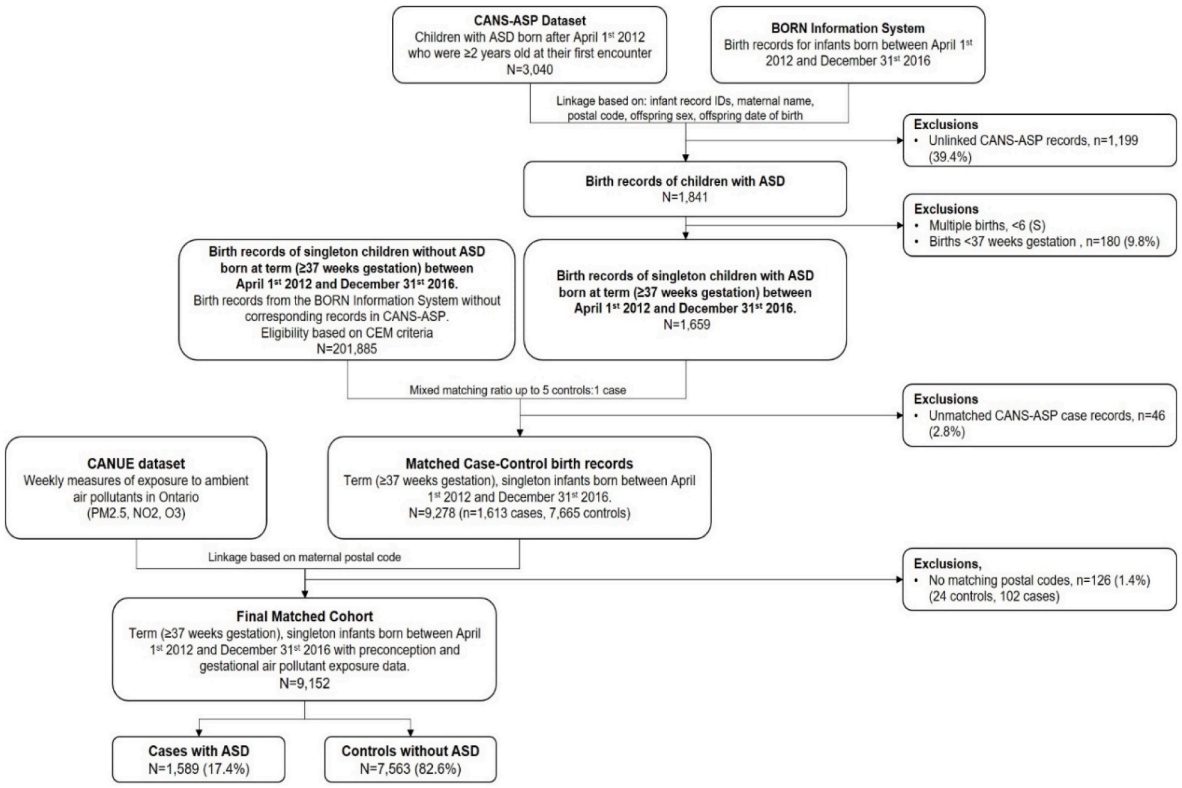


Fig. 1. Cohort creation strategy.

difference ≥0.10 (Guy et al., 2021). Summary statistics were also used to characterize the distribution of NO_2, O_3 and $PM_{2.5}$ concentrations over the study period, and Pearson's coefficients were used to quantify the correlation between air pollutants across trimesters.

The association between exposure to the specified air pollutants and ASD was estimated as adjusted odds ratios (OR) with 95% confidence intervals (Cis), applying conditional logistic regressions that accounted for multi-pollutant exposure. Six base models were run to estimate the associations for six exposure time window (T0-T5), which were required to examine time-varying associations. In addition to covariate matching, all models accounted for maternal age and infant birth season. Additional details on each model are provided in eAppendix 1.

For additional comparison of our findings against published studies applying distributed lag non-linear models (DLNMs) (Gasparrini, 2014) to investigate finer exposure windows, we applied DLNMs to study the effect of weekly exposures to each air pollutant on ASD, controlling for maternal age at delivery and birth season. We modelled each pollutant by using 1 to 9 knots and picked the best fit model using the Akaike Information Criterion (AIC), which is useful in comparison of multiple models for the same data set (Bozdogan, 1987). All models were conducted using complete cases analysis.

2.6. Sensitivity and additional analyses

Sensitivity analyses were conducted to assess model-dependency, i.e. the robustness of our findings across different models, in four ways. First, we assessed the association between each air pollutant and ASD using multi-pollutant conditional logistic regression models with additional adjustment for covariates that exhibited imbalance between cases and controls: mode of delivery (vaginal delivery vs caesarean section), maternal residential setting at delivery (urban vs rural), and infant sex (male vs female). Second, we examined the weekly exposures continuously from the pre-conception until the delivery for each air pollutant, while controlling for maternal age and birth season. Third, the weekly models were run with additional adjustment for mode of delivery, rurality, and infant sex. Finally, all base and sensitivity models were repeated using unconditional logistic regression.

All analyses were conducted using two statistical software: SAS version 9.4 (SAS Institute, Cary, North Carolina, USA) and R version 4.2.0 (R Foundation for Statistical Computing, Vienna, Austria) with a few R packages including dlnm (Gasparrini, 2014), lme4 (Bates et al., 2015), coxme (Therneau, 2022), medflex (Steen et al., 2017), dplyr (Wickham et al., 2022), and openxlsx (Schauberger et al., 2022).

2.7. Ethics and data privacy

This study was approved by the Ottawa Health Science Network Research Ethics Board (#20190660-01K), and the Health Canada Research Ethics Board (#2019-024H).

3. Results

3.1. Sample characteristics

Fig. 1 summarizes the strategy used for data linkage and cohort creation. The CANS-ASP dataset included records from 3040 children with ASD who were born after April 1st, 2012, and who were at least 2 years of age at their first encounter. After linkage with the BORN Ontario and the CANUE datasets, and application of exclusion criteria, the final matched cohort included 9152 children between the ages of 2 and 6 years; 1589 cases with ASD (17%) and 7563 controls without ASD (83%).

Maternal and infant baseline characteristics are detailed in Table 1. Cases and controls were balanced (SMD <0.1) for the following maternal covariates: age, neighbourhood education and household income quartiles, race/ethnicity, self-reported substance use in pregnancy, pre-existing health and mental health conditions (all SMD <0.1). Cases and controls also had similar infant characteristics including birth

year, birth season, birthweight, and gestational age at birth. Compared to controls, ASD cases were more likely to be born in urban communities (98% for cases vs 94% for controls, SMD = 0.2), to have been delivered by Caesarean section (33% vs 28%, SMD = 0.1), and to be male infants (79% vs 52%, SMD = 0.59). Accordingly, these covariates were adjusted for in subsequent sensitivity analyses.

3.2. Air pollutant concentrations and correlations

Descriptive statistics for ambient air pollution concentrations, including mean, standard deviation, median and IQR for each exposure period assessed are summarized in Table 2. Air pollutants showed little variation across the exposure periods examined; mean concentrations for the entire prenatal window (T5) for NO_2, O_3 and $PM_{2.5}$ were 11.9 ± 4.7 ppb; 47.4 ± 4.2 ppb and 7.9 ± 1.3 μg/m^3, respectively. There was positive moderate correlation between NO_2 and $PM_{2.5}$ across all prenatal windows (Pearson's correlation coefficient ranged from 0.41 in T0 to 0.53 in T5). In contrast, O_3 demonstrated weak and mixed correlations with the other two pollutants: negative correlations with NO_2 ranged from −0.34 in T5 to −0.26 in T1 but positive correlations with $PM_{2.5}$ ranged from 0.09 in T4 to 0.14 in T1. A summary of correlation coefficients between NO_2, O_3 and $PM_{2.5}$ across all prenatal windows examined in our analysis is available in eTable 1.

3.3. Associations between air pollutant exposure and ASD in children

3.3.1. Multi-pollutant conditional logistic regression models

Table 3 summarizes the OR and 95%CI per IQR change for each air pollutant obtained from multi-pollutant conditional logistic regression models. Overall, prenatal NO_2 exposure was positively and significantly associated with ASD in children: OR 2.1 (95%CI 1.8–2.3) per 7.3 ppb change for T0, OR 2.2 (95%CI 2.0–2.5) per 7.7 ppb change for T1, OR 2.2 (95%CI 1.9–2.5) per 7.5 ppb change for T2, and OR 2.1 (95%CI 1.9–2.4) per 7.0 ppb change for T3. In contrast, multi-pollutant base models returned weak or no association between prenatal exposure to ambient O_3 and $PM_{2.5}$, and ASD.

3.3.2. Single-pollutant distributed lag non-linear models (DLNM)

Fig. 2 illustrates the association between ASD and prenatal ambient air pollutants exposure by gestational week from the single-pollutant DLNMs across the entire study period from 13-weeks preconception to delivery. Lag weekly exposures to NO_2 during the pre-conception and gestational periods were positive and significant: per IQR increase, OR 1.2 (95%CI 1.1–1.2) for the pre-conception period and OR 2.5 (95%CI 2.2–2.8) for the gestational period. The magnitude of association reached a nadir in early-to-mid pregnancy and decreased towards delivery.

The lag pre-conception and gestational ORs for O_3 were negative overall, but only achieved statistical significance during the gestational period with an OR of 0.78 (95%CI 0.69–0.89) per IQR increase.

As with NO_2, the lag-response associations for $PM_{2.5}$ were positive and significant (OR 1.2, 95%CI 1.1–1.2 per IQR increase for the pre-conception period; OR 2.2, 95%CI 1.8–2.7 per IQR increase for the gestational period). Findings mirrored those for NO_2 with the association between $PM_{2.5}$ and ASD being strongest early-to-mid pregnancy, and the magnitude of association diminishing toward delivery.

3.3.3. Sensitivity analyses

Further adjustment for additional confounders including mode of delivery, residential setting and infant sex in our models did not substantively change our findings (Table 4). Cesarean delivery and male sex were positively and significantly associated with ASD, whereas maternal residence in rural communities during the prenatal period had an inverse association with ASD.

Single pollutant models for NO_2 and $PM_{2.5}$ that accounted for air pollutant exposure across each pre-conception and gestational week yielded similar findings to those from the DLNM (eTables 2 and 3). Briefly, NO_2 and $PM_{2.5}$ were positively and significantly associated with ASD, whereas O_3 showed a weak inverse association with ASD. These findings remained when adjusting for maternal age and birth season (eTable 2), and with additional adjustment for infant sex, mode of delivery and type of residential community (eTable 3). Results did not qualitatively differ in analyses using unconditional logistic regression (eTable 4-7; eFig. 1).

Table 2
Descriptive statistics for area-level air pollutant exposure.

	Exposure Period					
Air pollutant	T0	T1	T2	T3	T4	T5
NO_2, ppb						
Mean ±	11.9 ±	12.1 ±	12.0 ±	11.6 ±	11.9 ±	11.9 ±
SD	5.4	5.5	5.4	5.2	4.8	4.7
Median	11.7	12.1	11.8	11.4	12.4	12.6
(IQR)	(7.3)	(7.7)	(7.5)	(7.0)	(6.9)	(6.9)
O_3, ppb						
Mean ±	46.2 ±	46.2 ±	48 ±	48.4 ±	47.7 ±	47.4 ±
SD	9.4	9.5	9.4	8.9	4.7	4.2
Median	45.4	45.2	48.5	48.6	47.3	46.7
(IQR)	(15.3)	(15.8)	(15.3)	(14.2)	(5.8)	(5.4)
$PM_{2.5}$, μg/m^3						
Mean ±	7.9 ±	7.9 ±	7.8 ±	7.8 ±	7.8 ±	7.9 ±
SD	1.7	1.7	1.6	1.6	1.3	1.3
Median	7.9	7.8	7.7	7.8	8.0	8.1
(IQR)	(2.2)	(2.1)	(1.8)	(1.8)	(1.5)	(1.4)

IQR, interquartile range (difference between 75th and 25th percentile); ppb, parts per billion; SD, standard deviation; T0, 13-weeks preconception to conception; T1, 1st trimester; T2, 2nd trimester; T3, 3rd trimester; T4, conception to delivery; T5, 13-weeks preconception to delivery.

4. Discussion

In this matched case-control sample of nearly 10,000 children born in Ontario, Canada, we found a significant and consistent association between prenatal NO_2 exposure and ASD. Findings from multi-pollutant conditional logistic regression models demonstrated that maternal exposure to NO_2 was associated with increased odds of ASD both before (13 weeks prior to conception) and during pregnancy. These findings were supported by those from single-pollutant DLNMs, sensitivity analyses after adjustment for additional covariates, and unconditional logistic regression models. Unlike NO_2, the association between prenatal exposure to O_3 and $PM_{2.5}$, and ASD was largely non-significant and inconsistent, as the magnitude and direction of association varied depending on the prenatal exposure time windows assessed and various models tested.

The incidence of ASD has increased over time worldwide (Zeidan J et al., 2022), but it did not affect this study findings for three reasons. First, we included the year of birth as a matching variable as described in Study design and Table 1, which demonstrates balance between cases and controls. This means the birth year was controlled in this study. Second, the ASD cases in this study were children between 2 and 6 years old, and there was no noticeable increased prevalence of ASD (Table 1). Third, As a sensitivity analysis, we applied unconditional logistic regression model, including the birth year as a covariate. The results are reported in eTable 4 (supplementary materials, page 14–17). Overall, the birth year (each year) exhibited a null association with ASD, indicating no effect from the birth year.

Biological mechanism between air pollution and ASD: Hypotheses linking prenatal air pollution exposure with ASD in children have emerged as a topic of great interest over the last decade. Possible mechanisms underlying the biological plausibility of such a relationship are unclear, but there is sufficient evidence demonstrating that air pollutants can accumulate within the placenta (Bongaerts et al., 2022; Liu et al., 2021) are associated with diminished placental growth and

Table 3
Association between NO_2, O_3, $PM_{2.5}$ and covariates with offspring risk of ASD across prenatal windows of exposure.

Predictor	Exposure Period, OR (95%CI)					
	T0	T1	T2	T3	T4	T5
NO_2, per IQR increase						
T0	2.1 (1.8–2.3)	-	-	-	-	0.99 (0.65–1.5)
T1	-	2.2 (2.0–2.5)	-	-	1.7 (1.3–2.2)	1.6 (1.0–2.4)
T2	-	-	2.2 (1.9–2.5)	-	1.0 (0.73–1.4)	1.0 (0.69–1.5)
T3	-	-	-	2.1 (1.9–2.4)	1.3 (1.0–1.7)	1.4 (1.0–2.1)
O_3, per IQR increase						
T0	0.79 (0.66–0.94)	-	-	-	-	0.76 (0.55–1.0)
T1	-	0.83 (0.69–1.0)	-	-	0.84 (0.68–1.1)	0.93 (0.70–1.2)
T2	-	-	0.84 (0.71–1.0)	-	0.80 (0.64–1.0)	0.77 (0.59–1.0)
T3	-	-	-	0.85 (0.71–1.0)	0.85 (0.70–1.0)	0.99 (0.74–1.3)
$PM_{2.5}$, per IQR increase						
T0	1.1 (0.97–1.2)	-	-	-	-	1.1 (0.91–1.2)
T1	-	1.0 (0.94–1.1)	-	-	1.0 (0.90–1.1)	1.0 (0.88–1.2)
T2	-	-	1.0 (0.99–1.1)	-	1.1 (1.0–1.2)	1.1 (0.97–1.2)
T3	-	-	-	1.1 (0.99–1.1)	1.1 (0.96–1.2)	1.1 (0.94–1.2)
Maternal age, years	1.0 (0.99–1.1)	1.0 (0.99–1.1)	1.0 (0.99–1.1)	1.0 (0.99–1.1)	1.0 (0.99–1.1)	1.0 (0.99–1.1)
Birth Season						
Spring (Mar–May)	ref	ref	ref	ref	ref	ref
Summer (June–Aug)	0.79 (0.64–0.98)	0.48 (0.39–0.59)	0.80 (0.65–0.98)	1.4 (1.2–1.7)	0.79 (0.60–1.0)	0.76 (0.57–0.99)
Fall (Sept–Nov)	0.51 (0.39–0.67)	0.53 (0.44–0.65)	1.6 (1.2–2.1)	1.5 (1.3–1.9)	0.99 (0.71–1.4)	0.88 (0.62–1.3)
Winter (Dec–Feb)	0.54 (0.43–0.67)	0.97 (0.77–1.2)	1.7 (1.4–2.2)	0.91 (0.72–1.2)	1.1 (0.78–1.4)	1.0 (0.73–1.4)

OR, adjusted odds ratio; CI, confidence interval; T0, 13-weeks preconception to conception; T1, 1st trimester; T2, 2nd trimester; T3, 3rd trimester; T4, conception to delivery; T5, 13-weeks preconception to delivery.

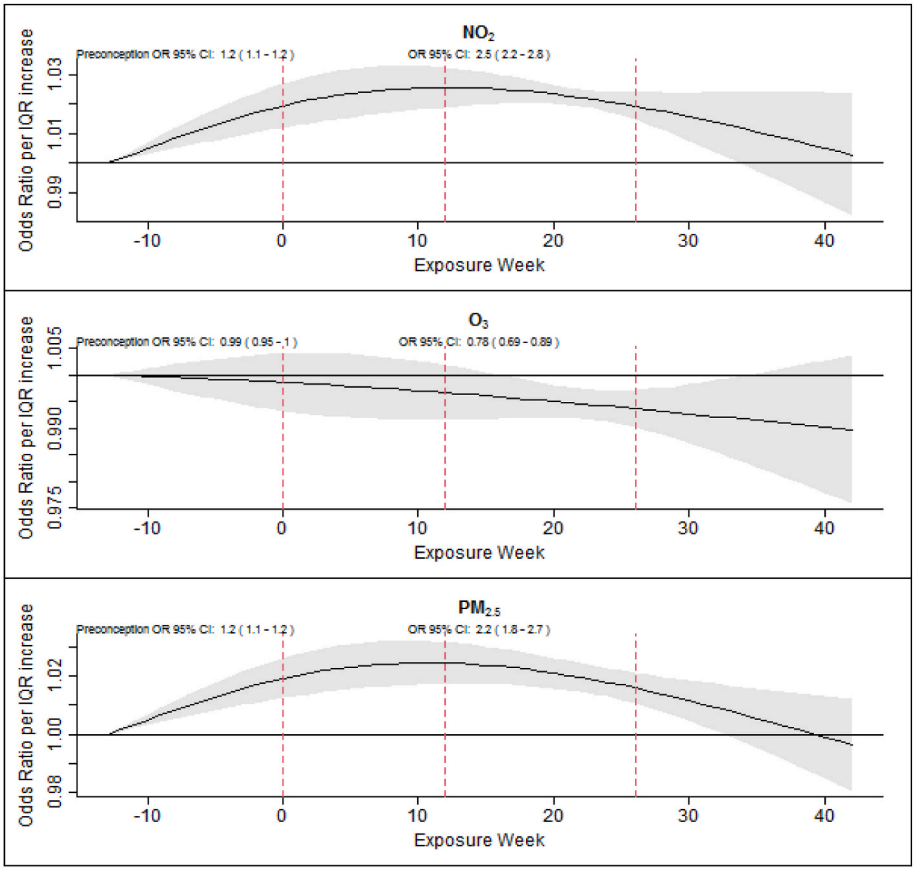


Fig. 2. Distributed lag non-linear models for NO_2, O_3, and $PM_{2.5}$. Presented models are based on AIC: NO_2, 2 knots; O_3, 2 knots; $PM_{2.5}$, 2 knots. Vertical dashed lines demarcate the following events, from left to right: conception, end of the 1st trimester, end of the 2nd trimester. Odds ratios (OR) are presented for both preconception and gestational.

function (Saenen et al., 2019b; van den Hooven et al., 2012b; Liu et al., 2018) and can cross into fetal circulation (Bongaerts et al., 2022). Networks of candidate genes linked to ASD include those implicated in brain development and neuronal signaling, which may well be susceptible to the inflammatory effects of air pollutants (Costa et al., 2017). Pollution-mediated increases in peripheral oxidative stress and inflammation in the central nervous system have been shown to produce a range of lasting neurocellular effects that may disrupt cognitive,

Table 4
Association between NO_2, O_3, $PM_{2.5}$ and covariates with offspring risk of ASD across prenatal windows of exposure with additional adjustment for mode of delivery, type of residential community and infant sex.

Predictor	Exposure Period, OR (95%CI)					
	T0	T1	T2	T3	T4	T5
NO_2, per IQR increase						
T0	2.1 (1.8–2.3)	-	-	-	-	1.0 (0.67–1.6)
T1	-	2.2 (2.0–2.5)	-	-	1.7 (1.3–2.2)	1.5 (0.95–2.4)
T2	-	-	2.1 (1.9–2.4)	-	1.0 (0.73–1.5)	1.1 (0.70–1.6)
T3	-	-	-	2.1 (1.8–2.4)	1.3 (0.97–1.6)	1.3 (0.86–1.9)
O_3, per IQR increase						
T0	0.79 (0.65–0.95)	-	-	-	-	0.81 (0.58–1.1)
T1		0.84 (0.70–1.0)	-	-	0.86 (0.69–1.1)	0.92 (0.68–1.2)
T2	-	-	0.81 (0.67–0.98)	-	0.77 (0.61–0.97)	0.75 (0.57–0.99)
T3	-	-	-	0.81 (0.67–0.98)	0.81 (0.66–1.0)	0.91 (0.66–1.2)
$PM_{2.5}$, per IQR increase						
T0	1.1 (0.94–1.2)	-	-	-	-	1.0 (0.90–1.2)
T1	-	1.0 (0.91–1.1)	-	-	0.98 (0.87–1.1)	1.0 (0.87–1.2)
T2	-	-	1.0 (0.93–1.1)	-	1.1 (1.0–1.2)	1.1 (0.97–1.2)
T3	-	-	-	1.0 (0.94–1.1)	1.1 (0.95–1.2)	1.0 (0.98–1.1)
Maternal age	1.0 (0.98–1.1)	1.0 (0.98–1.1)	1.0 (0.99–1.1)	1.0 (0.99–1.1)	1.0 (0.98–1.1)	1.0 (0.98–1.1)
Birth Season						
Spring (Mar–May)	ref	ref	ref	ref	ref	ref
Summer (June–Aug)	0.77 (0.62–0.96)	0.47 (0.38–0.58)	0.81 (0.65–1.0)	1.4 (1.1–1.7)	0.79 (0.60–1.0)	0.77 (0.58–1.0)
Fall (Sept–Nov)	0.52 (0.39–0.69)	0.54 (0.44–0.67)	1.7 (1.3–2.2)	1.5 (1.2–1.8)	1.0 (0.72–1.5)	0.93 (0.64–1.4)
Winter (Dec–Feb)	0.52 (0.42–0.65)	0.92 (0.73–1.2)	1.7 (1.4–2.1)	0.84 (0.65–1.1)	0.98 (0.72–1.3)	0.95 (0.68–1.3)
Delivery mode						
Vaginal	ref	ref	ref	ref	ref	ref
Caesarean	1.2 (1.1–1.4)	1.2 (1.1–1.4)	1.2 (1.0–1.3)	1.2 (1.0–1.3)	1.2 (1.1–1.4)	1.2 (1.1–1.4)
Residential community type						
Urban	ref	ref	ref	ref	ref	ref
Rural	0.47 (0.32–0.7)	0.48 (0.32–0.72)	0.47 (0.31–0.70)	0.47 (0.31–0.70)	0.52 (0.34–0.80)	0.53 (0.34–0.81)
Infant sex						
Female	ref	ref	ref	ref	ref	ref
Male	3.6 (3.1–4.1)	3.5 (3.1–4.1)	3.6 (3.1–4.1)	3.6 (3.1–4.1)	3.6 (3.1–4.1)	3.6 (3.1–4.1)

OR, odds ratio; CI, confidence interval; ref, reference category; T0, 13-weeks preconception to conception; T1, 1st trimester; T2, 2nd trimester; T3, 3rd trimester; T4, conception to delivery; T5, 13-weeks preconception to delivery.

behavioural, and emotional development (Costa et al., 2019).

Single-vs. multi-pollutant models for ASD: In our analyses, maternal exposure to NO_2 in pregnancy was a significant contributor to ASD risk, after accounting for co-pollutant exposures, sociodemographic and clinical covariates. The cumulative odds for ASD associated with weekly NO_2 exposure alone across pregnancy was also significant, although somewhat higher than what has been reported elsewhere. In a meta-analysis of five studies reporting NO_2 specific associations with ASD, Chun et al. found a marginally significant increase in ASD (OR 1.06, 95%CI 1.01–1.11 per 10 $\mu g/m^3$ increase) (Chun et al., 2020). Similarly, Dutheil et al. reported a 2% increased risk of ASD for each 10 $\mu g/m^3$ increase in NO_x (OR 1.02, 95%CI 1.01–1.03) (Dutheil et al., 2021). These findings are in contrast, however, to those of a meta-analysis published in 2016 and more recent population-based retrospective cohort study that observed no significant association between prenatal NO_2 exposure and ASD in children (Flores-Pajot et al., 2016; Rahman et al., 2022). Although several studies have linked maternal O_3 (Becerra et al., 2013; Kim et al., 2017; McGuinn et al., 2020) and $PM_{2.5}$ (Becerra et al., 2013; Raz et al., 2015; Volk et al., 2013) exposure during pregnancy to subsequent ASD diagnoses in children, findings from multi-pollutant models in the current analysis would suggest otherwise. Comparable literature in this area predominantly focuses on single-pollutant effects however, and inconsistency in reported findings may be attributable to differences in analytical approaches. DLNM and single-pollutant conditional regression models in our analysis revealed a significant positive association between prenatal $PM_{2.5}$ exposure and ASD. The reason for the discrepancy between these findings and those from our multi-pollutant conditional logistic regression models remains unclear. Multi-pollutant models better reflect real-world exposures and outcomes however, and perhaps provide more insight into the relationship between prenatal air pollutant exposure and later risk of ASD. Data from meta-analyses are also inconsistent. In a meta-analysis of 23 studies, Lam et al. found evidence for an association between prenatal $PM_{2.5}$ and PM_{10} exposure and ASD (adjusted OR 2.32 (95%CI 2.15, 2.51) per 10-$\mu g/m^3$ increase in $PM_{2.5}$ and OR 1.07 (95%CI 1.06, 1.08) per 10-$\mu g/m^3$ increase in PM_{10}) (Lam et al., 2016). In another analysis by Flores-Pajot et al., meta-estimates and weights for prenatal exposure to $PM_{2.5}$ and PM_{10} were not statistically significant, but analyses for O_3 yielded a risk ratio of 1.50 (95%CI 1.01, 1.10) per 10 ppb increase (Flores-Pajot et al., 2016). Associations within each trimester of pregnancy were largely positive, but summary measures generally did not reach statistical significance (Flores-Pajot et al., 2016). More recently, meta-analyses conducted by our research team demonstrated positive and significant associations of prenatal $PM_{2.5}$ exposure with ASD, based on both fixed effects and random effects models (Chun et al., 2020). Pooled estimates for PM_{10} and O_3 were not significant and time-varying associations across pregnancy trimesters were inconsistent.

Maternal exposure to NO2 and ASD in young children: Both NO2 and PM2.5 are harmful pollutants, sharing some sources such as burning of fossil fuels and biofuels. While PM2.5 comes from more various sources such as traffic, industrial plants, and wildfires, NO2 comes largely from traffic and industrial plants and is a key precursor in formation of ground-level ozone and also contributes to the formation of PM2.5 in the air (NASA, 2024). Among the three air pollutants (PM2.5, NO2, and ozone), PM2.5 is the one most frequently reported to be associated with ASD in meta-analyses (Lam et al., 2016; Chun et al., 2020). However, this study found inconsistent PM2.5-ASD associations, which were significant only from a single-pollutant model (not multi-pollutant model). In contrast, NO2-ASD associations were observed consistently from both single- and multi-pollutant models. The difference in association with ASD between NO2 and PM2.5 can be explained statistically for two reasons: correlation and range of concentrations (or variations). First, the correlation between NO2 and

PM2.5 were overall moderate, between 0.3 and 0.5 (eTable 1, Supplementary) and NO2 a precursor of PM2.5. Second, NO2 concentrations were higher with larger variations (as indicated by SD or IQR) than PM2.5 with smaller variations (Table 2). This implies that NO2 has more statistical power (rejecting the null association) than PM2.5. Combining these two observations, it is expected that PM2.5 could lose its association with ASD in the multi-pollutant model for PM2.5 and NO2, which was observed in this study.

A variety of pathophysiological mechanisms linking NO2 (or traffic-related pollution more generally) and ASD have been hypothesized through dramatic increase in nitric oxide levels in ASD mouse models and nitrosative stress biomarkers in the plasma of low-functioning ASD patients (Gong et al., 2017; Guxens et al., 2016; Hine et al., 1970; Tripathi et al., 2023), and through oxidative stress, neuroinflammation, epigenetic alterations, and impaired signal transduction (Persico and Merelli, 2014). However, further research is needed to establish causality.

Exposure time windows (time-varying associations across pregnancy): A handful of studies reporting on trimester-specific associations with ASD have yielded inconsistent results, although again, the majority of these have relied on single-pollutant models that average exposures over trimesters or the entire pregnancy period (Dutheil et al., 2021). Even fewer have investigated exposures prior to pregnancy (McGuinn et al., 2020; Raz et al., 2015; Talbott et al., 2015). The results of our own analyses revealed no notable variation in the magnitude of association across each trimester when the effects of NO2 were modelled with O_3 and $PM_{2.5}$ by conditional logistic regression. Here, NO_2 was a statistically significant and consistent contributor to ASD risk before conception, and in each trimester of pregnancy, whereas O_3 and $PM_{2.5}$ were not. It was only through a single-pollutant DLNM approach that we observed differences in fetal susceptibility by gestational age of air pollutant exposure. These findings suggest that pre-conception and early-to-mid pregnancy are periods during which maternal exposure to both NO_2 and $PM_{2.5}$ were strongly associated with ASD in children. The magnitude of this association for both these pollutants reached a nadir in the 1st and 2nd trimesters and diminished toward delivery. However, a population-based nested case-control study in Israel reported higher odds of ASD for postnatal exposure to NO_2than prenatal exposure (Raz et al., 2018). The variation in the observed patterns and magnitude of the exposure-outcome association across the prenatal period according to the modelling approach used underscores the need for careful consideration in the choice of study design and analytical methods, and in the interpretation of published findings.

For critical exposure time windows, this study found stronger associations during early-to-mid pregnancy than late-pregnancy. However, mechanistic studies of NO2-induced fetal impacts are limited and inconsistent with maternal exposure. Fussell et al. (2024) reviewed underlying mechanisms between air pollution and adverse health outcomes. Several studies investigated, in particular, placental vascularization and function based on maternal exposures. They found that NO2 exposure associated with increased incidence of bilateral uterine notching (van den Hooven et al., 2012b), with a lower mean artery pulsatility index (Hettfleisch et al., 2017; Cahuana-Bartra et al., 2022), with an increased uterine resistance index (Contreras et al., 2018), and with an increased resistance index of the left uterine artery (Ouidir et a., 2021). These underlying mechanisms suggest adverse health effects of air pollution exposure on blood viscosity, endocrine disruption, fetal growth, hemodynamic endothelial function, inflammation, and oxidative stress (Kannan et al., 2006; Slama et al., 2008; Shah et al., 2011; Backes et al., 2013; Smarr et al., 2013; Feng et al., 2016; Carvalho et al., 2016; Stieb et al., 2019). NO2 can induce morphological changes that disrupt the flow of oxygen and nutrients across the placenta through these mechanisms. The observed early-to-mid pregnancy in this study may be associated with these mechanisms, but its link to ASD remains unknown.

The concomitant influence of multiple environmental exposures should also be considered. Future investigations should account for exposure to multiple pollutants (NO_x, O_3, PM_x, SO_2, CO), include granular measures of exposure (e.g., by gestational week rather than averaged over trimesters), measures of the built environment (e.g., walkability, green space, blue space, light pollution) and account for a broad range of sociodemographic and clinical covariates.

In addition to the trimesters, birth season was also examined in relation with ASD, and Attention-deficit/hyperactivity disorder (ADHD). Both ASD and ADHD are considered as neurological disorders. A multinational study on five European countries (Lee et al., 2019) reported birth seasonality in relation to ASD: highest risk of ASD for fall births (conceived in the winter) and lowest risk for spring birth (conceived in the summer). They hypothesized that seasonality was explained by solar radiation levels (sunlight levels) during the conception and trimester 1. The solar radiation level is generally low in the winter and high in the summer, so they suggested an inverse correlation between sunlight levels and ASD. However, another study on upstate New York counties reported the opposite result that spring birth (April–May), compared with summer birth (June–August), had significantly more ADHD symptoms (Zhang et al., 2018). This study examined not only time-varying exposures by trimester but also birth season (Table 3) and found their interactions as follows: overall summer birth (June–August), compared with spring birth (March–May), had less ASD for Trimesters 0–2, but more ASD for Trimester 3. This finding cannot be explained by the sunlight levels at conception and needs further investigation on seasonality of each trimester.

Study limitations: There are limitations to our analysis. First, this study may not have a representative sample of all children with ASD in Ontario, or elsewhere for two reasons related to the database. The CANS-ASP dataset is limited to children with ASD receiving the ABA services reported from the 13 agencies in the province. Thus, children with ASD accessing privately funded care and those outside of the 13 agencies may not be represented by our sample cases. This might have resulted in unmeasured bias in the ORs reported in this study. Second, our analyses did not account for phenotype heterogeneity. Information on ASD disease severity was not available for these analyses, and therefore we were unable to explore the exposure-outcome association by outcome severity. Third, changes in residential addresses could not be confirmed for the study sample, and therefore our analyses assume that postal codes remained consistent throughout the exposure period. However, another Canadian cohort study on maternal exposures reported that absence of residential moving history returned comparable estimated associations for both weekly and trimester exposures, after examining single location exposures versus averaged exposures based on moving histories (Johnson et al., 2022). Fourth, although the CEM method for matched case-control design established the desirable balance between cases and controls for the covariates used in the study but not for mediators and confounders that were unavailable in our dataset. Notable factors not accounted for in our analyses include genetic and other prenatal environmental influences that may influence offspring risk of ASD. The recurrence of ASD in children with older affected siblings reportedly ranges from 4 to 18% (Hansen et al., 2019; Palmer et al., 2017) however, our small sample size precluded a sibling analysis (~1.6% of cases had sibling in the sample). Fifth, other air pollutants such as CO, SO_2 and other PM_x may also be associated with ASD but were not assessed in these analyses. In addition, the reductive effect of green space environments on air pollutant concentrations,(Diener and Mudu, 2021) and their beneficial health effects are also well-described (Kondo et al., 2018; McCormick, 2017). Further investigation into the role of maternal green space exposure during pregnancy in mitigating offspring risk of ASD is warranted.

Study strength: Notwithstanding the limitations, this study has several strengths. A significant strength of this study is our use of the CANS-ASP dataset to identify children who received ASD diagnoses from a health professionals including pediatric psychiatrist or psychologist. Other studies frequently rely on diagnostic codes and case finding

algorithms, which despite their merit, are subject to misclassification bias. Our findings are strengthened by our use of a well-characterized sample of cases and controls for which extensive maternal and infant covariate data were available. The population-based BORN Information System also enabled identification of many matched controls with which ASD cases could be compared. Further, our analyses tested exposure-outcome associations using models that accounted for both multiple and single pollutant exposure, and the availability of weekly air pollution data made it possible for us to examine critical windows of exposure including 13-week pre-conception, and across each trimester of pregnancy.

This study adds to mounting evidence suggesting that air pollutants may be key environmental factors to the pathogenesis of ASD. We show that maternal exposure to higher concentrations of NO_2 both before and during pregnancy may be associated with greater odds of ASD in children. Further investigation using population-based cohorts to assess the combined influence of air pollutants on fetal and child neurodevelopment is warranted. Importantly, multi-pollutant models that account for co-exposure to dominant criteria air pollutants and continuous measures of exposure across the prenatal period should be considered.

5. Conclusion

This study found significant and consistent associations between maternal exposures to NO_2 and ASD in children, but not time-varying effects. Ambient NO_2 exposure during the study period, pre-pregnancy (13-week prior to conception) and pregnancy periods, increased the ASD risk in children, whereas maternal exposures to O_3 and $PM_{2.5}$ remains unclear. Our extensive analyses suggest no critical exposure time window that increased the association across each trimester when the effects of NO2 were modelled along with the co-pollutants. For air pollution related neurodevelopment disorders in young children, further investigations are warranted for combined effects of multi-pollutant exposures and for time-varying effects of pre-pregnancy, prenatal and postnatal exposures.

Funding

This study was funded by a research grant from Health Canada, “Addressing Air Pollution Horizontal Initiative” [Funding reference number, AAPHI-810626; Principal Investigator, Shin HH].

Data sharing

The data analyzed for this study are held securely at the prescribed registry, BORN Ontario. Data sharing regulations prevent these data from being made available publicly due to the personal health information in the datasets. Enquiries regarding BORN Ontario data, including the de-identified study dataset, must be directed to BORN Ontario (Science@BORNOntario.ca). Nitrogen dioxide data, calculated ozone metrics and PM2.5 metrics indexed to DMTI Spatial Inc. postal codes, were provided by CANUE (Canadian Urban Environmental Health Research Consortium).

CRediT authorship contribution statement

Malia SQ. Murphy: Writing – review & editing, Writing – original draft, Resources. **Kasim E. Abdulaziz:** Visualization, Validation, Software, Formal analysis, Data curation. **Éric Lavigne:** Writing – review & editing, Resources, Methodology, Formal analysis. **Erica Erwin:** Visualization, Validation, Software, Resources, Formal analysis, Data curation. **Yanfang Guo:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Alysha LJ. Dingwall-Harvey:** Writing – review & editing, Data curation. **David Stieb:** Writing – review & editing, Supervision, Formal analysis. **Mark C. Walker:** Writing – review & editing, Supervision, Formal analysis. **Shi Wu Wen:** Writing – review & editing, Supervision, Formal analysis. **Hwashin Hyun Shin:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgements

This study was supported by an Addressing Air Pollution Horizontal Initiative research grant from Health Canada [Funding reference number AAPHI-810687]. The authors would like to thank Dr. Lise Bisnaire, of BORN Ontario, for her guidance on the use of the Ontario Ministry of Children and Youth Services’ Child and Adolescent Needs and Strengths – Autism Spectrum Profile (CANS-ASP) dataset.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envres.2024.119706.

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